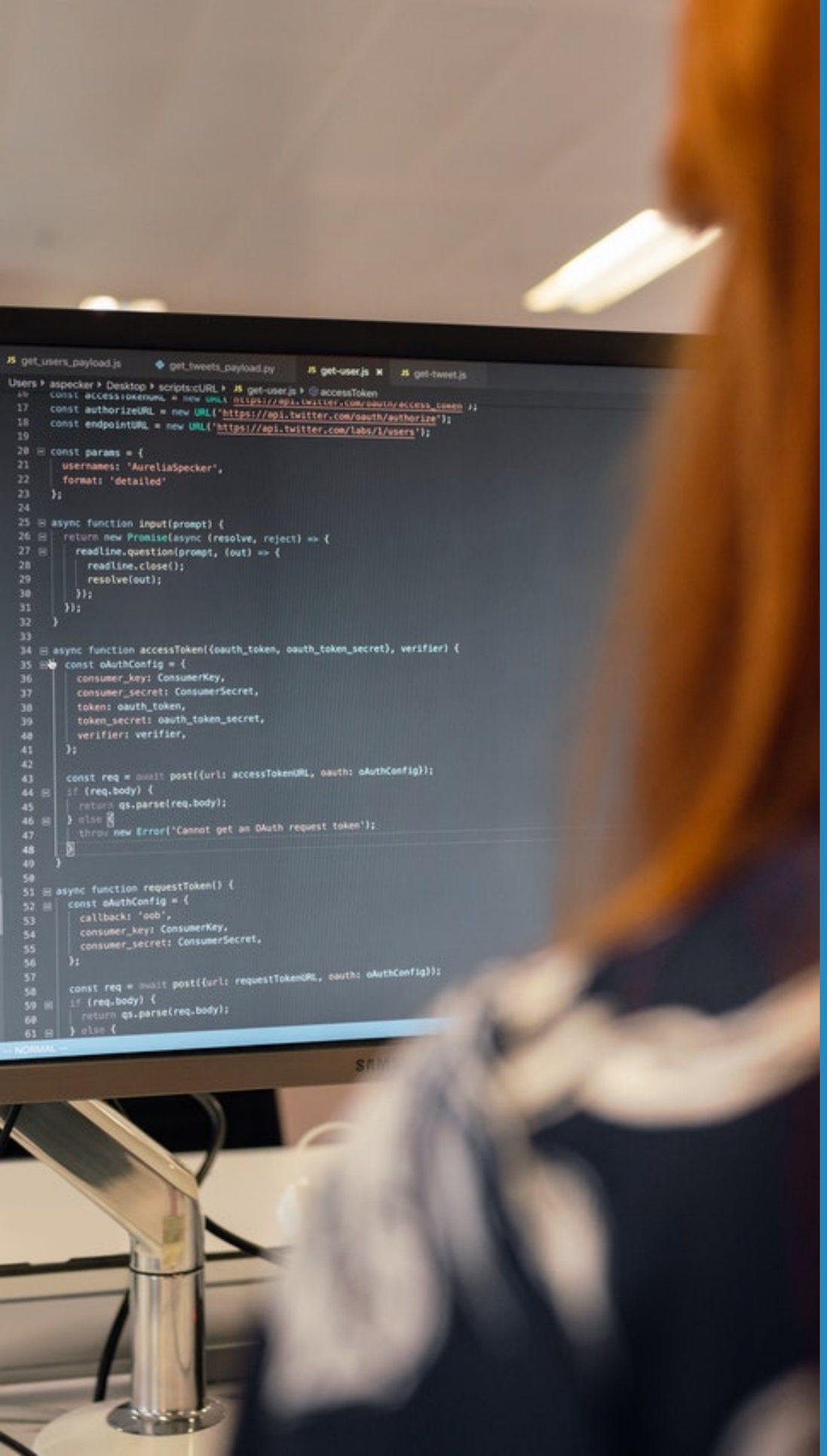
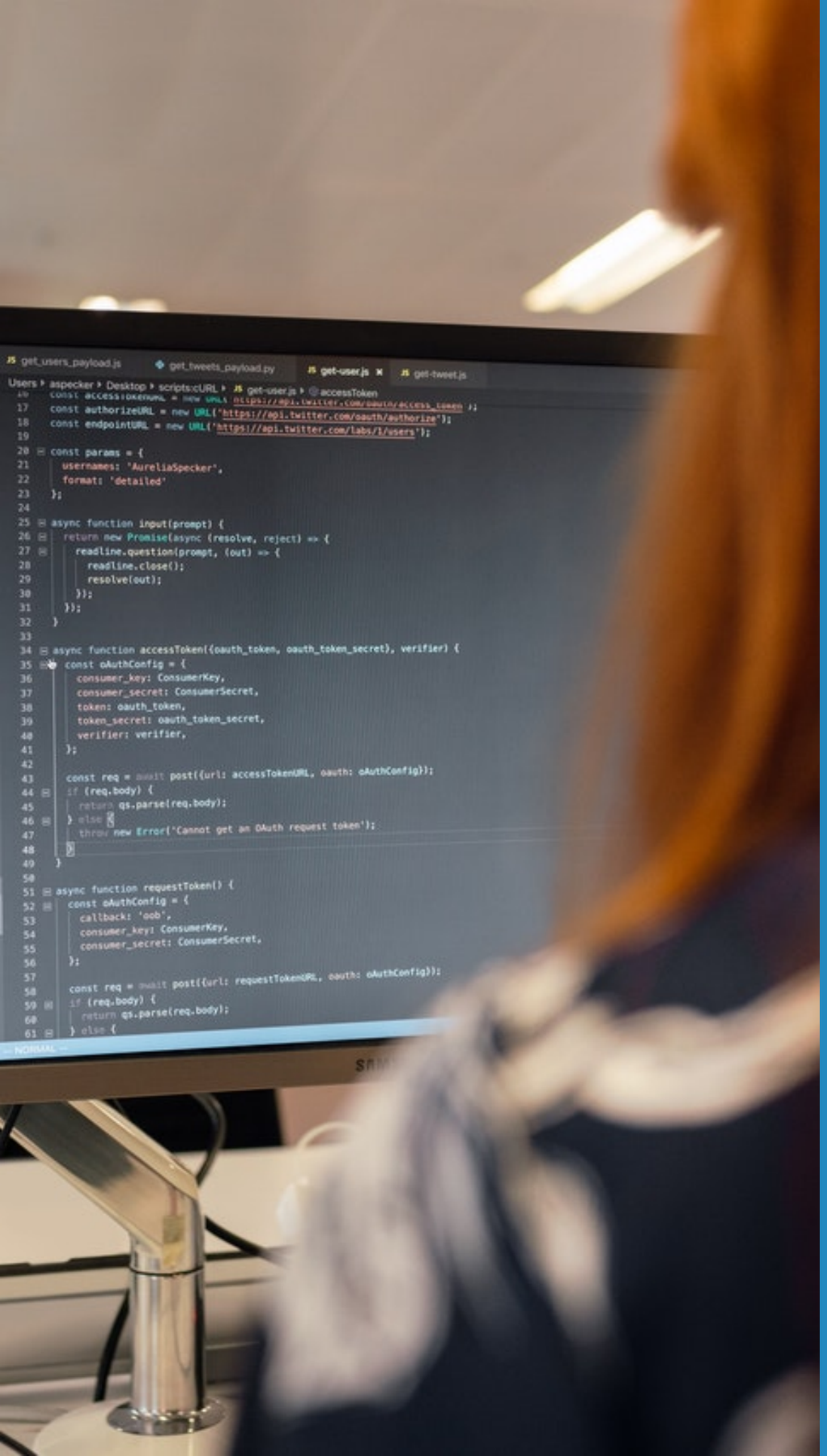
COMP1006 - WEEK FOUR

WORKING WTH FORMS & FORM VALIDATION



TODAY'S SCHEDULE

1. Check In & Review
2. Review Lab Two
3. Working with Form Data
4. Form Validation
5. A Little Bit About Bootstrap
6. Lab Three
7. Summary, Task List, Next Week



LEARNING OBJECTIVES

1. Capture user data from an HTML form

2. Validate user supplied data using client-side and server-side techniques

3. Utilize Bootstrap web framework for UI design

WEEK THREE / LAB TWO

REVIEW

SYNTAX ERRORS

- prevent your application from running.
- they are the easiest to find and fix.
- PHP interpreter isn't able to parse the script, so it displays an appropriate error message.

RUNTIME ERRORS

- does not violate the syntax rules, but they cause the PHP interpreter to display errors.
- may or may not stop execution of the script (fatal errors)
- the error message that's displayed usually can help you to find and fix the error.

LOGIC ERRORS

- can come from many places, and they are often the most difficult to find and fix.
- Logic errors are statements that don't cause syntax or runtime errors
- produce the wrong result

COMMON SYNTAX ERRORS

- Misspelling keywords.
- Forgetting an opening or closing parenthesis, bracket, brace, or comment character
- Forgetting to end a PHP statement with a semicolon.
- Forgetting an opening or closing quotation mark.
- Not using the same opening and closing quotation mark.

VARIABLE NAMES

- Misspelling or incorrectly capitalizing a variable name.
- Using a keyword as a variable name.

DEBUGGING TOOLS

- error messages
- var_dump
- error_logs

WHY PROJECT STRUCTURE MATTERS

```
/project-root
  /includes
    header.php
    footer.php
/assets
  css/
index.php
```

- Separate concerns (logic vs presentation)
- Encourages reuse (DRY!)
- Prepares for larger apps
- **Hint: use include/ require**

LAB TWO REVIEW

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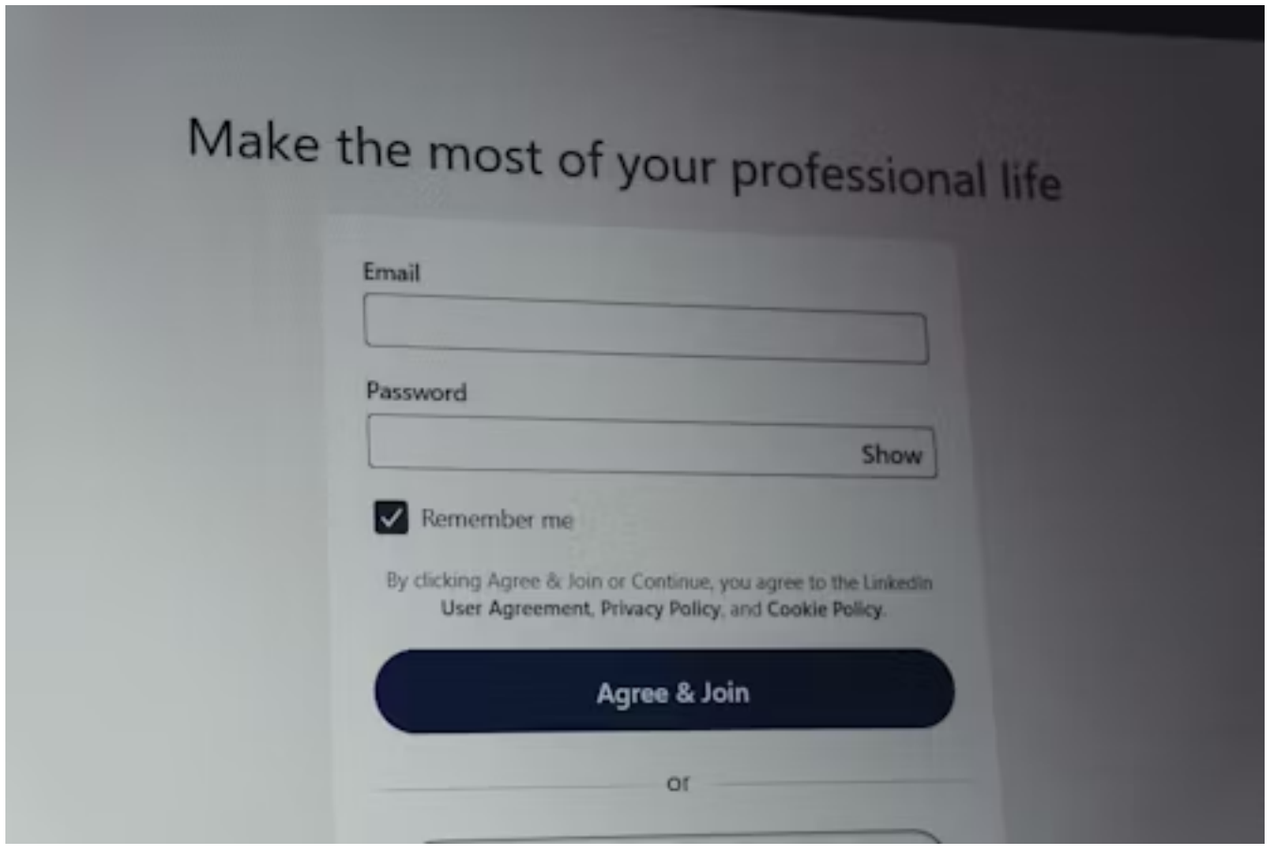
WORKING WITH FORM DATA

WORKING WITH FORM DATA – WHAT COULD GO WRONG ?



COMP1006 - WEEK FOUR

Re: Issues with the Order Form

Lisa Dough <lisa@bakeittillyoumakeit.com>
Today, 9:17 AM

Hi Team,

We're having some real headaches with the order form submissions! 🙈😉

- **Missing Information:** We're getting orders with no names, no phone numbers, or no addresses at all!
- **Weird Entries:** We're seeing "1234" for a name and "ASDF" as a phone number. Someone even put "Narnia" as an address! 🧥🌲❄️
- **Zero Quantities:** Some customers are submitting orders with ***zero items*** selected and nothing to bake! 💔😞

Plus! Could you have all order information sent to bitumi@gmail.com?
It seems like we've been missing them!

To: **bitumi@gmail.com**

Can you fix this ASAP? We can't keep guessing what people want, and it's causing chaos in the kitchen! 👩‍🍳

Thanks,
Lisa
Lisa Dough
Bake It Till You Make It Bakery🏠

An email was sent to **bitumi@gmail.com**

- Subject: New bakery order submission
- Attachments: Form data (txt file)

Reply Forward

WORKING WITH FORMS – ACTION & METHOD

1. Create a file called process.php. In that file, echo out a simple thank you message to the user. **What do you think should happen in process.php?**

2. Experiment changing the method in the form from post to get. **What do you notice?**

WORKING WITH FORMS – ACTION & METHOD

1. Create a file called process.php. In that file, echo out a simple thank you message to the user. **What do you think should happen in process.php?**

2. Experiment changing the method in the form from post to get. **What do you notice?**

THE <FORM> TAG: ACTION AND METHOD

```
<form action="process.php" method="post">
```

Action: where the form is sent

Method: how the information is sent

METHOD = GET

```
<form action="process.php" method="post">
```

- Form data is sent in the URL
- Data is visible in the address bar
- Data can be bookmarked or shared
- Best for:
  - search forms
  - filters
  - non-sensitive data

METHOD = POST

```
<form action="process.php" method="post">
```

- method="post"
- Form data is sent in the request body
- Data does not appear in the URL
- More secure than GET (but not encrypted)
- Best for:
  - orders
  - logins
  - forms with lots of data

ADD VALIDATION ON THE CLIENT SIDE

CLIENT-SIDE VALIDATION WITH HTML5 ATTRIBUTES

HOW TO ACCESS FORM DATA?

CHALLENGE : TRY TO COMPLETE PROCESS.PHP. THE SCRIPT SHOULD GRAB FORM DATA AND MAIL ORDER TO THE CLIENT (LISA) [WITHOUT USING AI.]

WHAT DID YOU LEARN?

WORKING WTH FORM DATA DEMO

PHP SUPERGLOBALS

1. Built in PHP variables that hold associative arrays
2. $_POST & $_GET
3. Access form data using name attribute as the key value

COMP1006 - WEEK FOUR

FORM VALIDATION

COMP1006 - WEEK FOUR

Emails Working, But Still Getting Blank Forms

Lisa Dough <lisa@bakeittillyoumakeit.com>
Today, 9:24 AM

Hi Team,

Good news! The emails with our order forms are coming in just fine. 🎉

But...we're still getting some submissions where the forms are completely blank. 😉 Can somebody still send us a form without filling out any of the fields?

I'm starting to get a littie concerned about the security of our form. Can we please look into this?

Thanks,
Lisa
Lisa Dough
Bake It Till You Make It Bakery🏠

Reply Forward

WHAT'S WRONG WITH OUR FORM?

VALIDATING DATA

- Need to ensure that we are receiving the right data
- Client side validation is for UX, server-side validation is for security

SANITIZING DATA

- Data supplied by the user is insecure
- Removing illegal characters from the data
- PHP provides a native filter extension that you can use to sanitize data and validate external input (filter_var, filter_input)

ADD VALIDATION ON THE SERVER SIDE

SERVER SIDE VALIDATION DEMO

COMP1006 - WEEK FOUR

A LITTLE MORE BOOTSTRAP

COMP1006 - WEEK FOUR

Thanks for Securing the Form!

Lisa Dough <lisa@bakeittillyoumakeit.com>
Today, 10:17 AM

Hi Team,

I just wanted to thank you for securing our online order form and making sure everything works as it should. It's looking great! 💗

I really appreciate you making it safer and preventing those blank orders from coming through. 🛡️ Now I can relax knowing our form is secure and we're getting complete, legit orders. 🧁

It really makes a big difference for our bakery. Thank you so much for all your hard work! 😊

One more thing: when you get a chance, could we make our form look a little nicer? We don't have a lot of dough to spend, but even a small makeover would be sweet. 🧁 

Thanks again,
Lisa Dough
Bake It Till You Make It Bakery🏠

Reply Forward

HOW COULD YOU IMPROVE THE LOOK OF THIS FORM WITHOUT IT COSTING THE CLIENT A LOT OF $$$

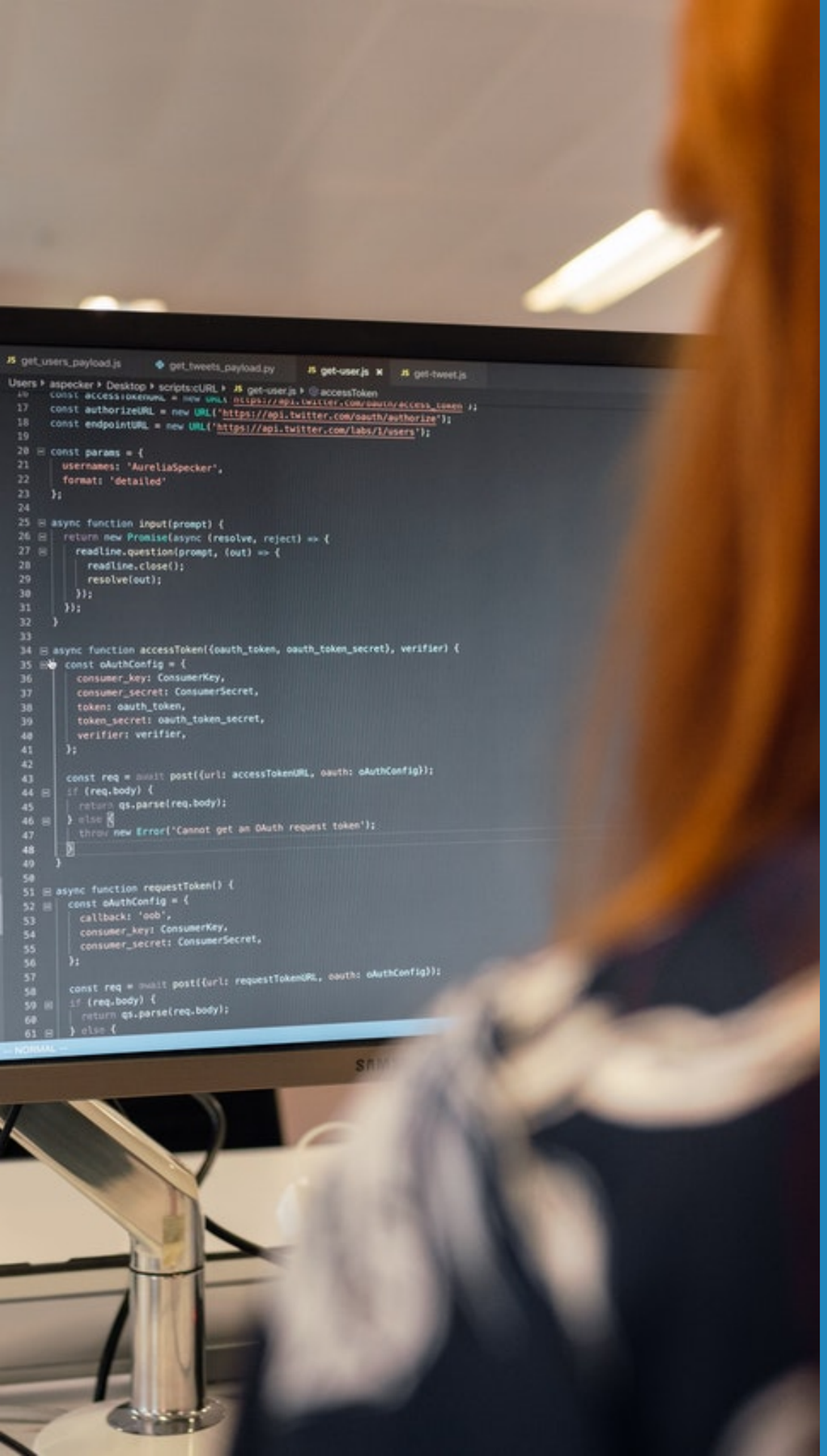
A LIL' BIT ABOUT BOOTSTRAP

- a framework for building responsive, mobile-first websites & applications
- Excellent for rapid development
- Easy to use & integrate
- Some drawbacks

BOOTSTRAP DEMO

LAB THREE

SUMMARY, TO DO & NEXT WEEK



SUMMARY

1. HTTP requests (get & post) and working with Superglobals
2. Form Validation - Client Side & Server Side
3. A Little Bit Of Bootstrap

TO DO THIS WEEK :

LAB THREE (DUE FRIDAY, JAN 30TH @ 11:59PM)

NEXT WEEK :

CREATE & READ